

An Internship Report
Of
Artificial Intelligence Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)



By

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Batch: 2026-27

CANDIDATE'S DECLARATION

I, ADITYA PRAKASH, do hereby declare that the internship report titled “**Artificial Intelligence Internship**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name: **ADITYA PRAKASH**

Roll No.: **25SCS1003005350**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

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Roll No.: **25SCS1003005350**

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3. INTERNSHIP COMPLETION CERTIFICATE



4. PROJECT DESCRIPTION

4.1 Introduction

The Artificial Intelligence Internship was designed to provide an introduction to the fundamental concepts, techniques, and applications of Artificial Intelligence and Machine Learning. The internship followed a structured learning approach covering Python for AI development, data preprocessing and visualization, supervised learning, model evaluation, neural networks, Convolutional Neural Networks (CNNs), Natural Language Processing (NLP), and Reinforcement Learning.

During the internship, theoretical concepts were supported by practical examples and implementations to develop an understanding of how AI techniques can be applied to real-world problems. As part of the practical component, a Content-Based Movie Recommendation System was developed using Python and machine learning techniques. The project provided an opportunity to apply concepts related to data handling, feature extraction, NLP, and similarity-based recommendation in a practical application.

4.2 Organization Profile

Codec Technologies is an IT consultancy, technology services, and educational training organization. The organization provides IT solutions including web and application development, cloud hosting, software deployment, infrastructure management, and digital transformation services.

Codec Technologies also provides corporate training and internship programs in various technical domains such as Artificial Intelligence, Data Science, Python Development, Cyber Security, Web Development, and Data Analytics. Its internship programs are designed to provide learners with structured training and practical exposure to industry-related technologies.

4.3 Problem Statement

Artificial Intelligence is a broad field involving various techniques such as machine learning, deep learning, natural language processing, and reinforcement learning.

Understanding these concepts requires not only theoretical knowledge but also practical exposure to data processing, model development, and evaluation.

The internship aimed to address the need for a structured introduction to these fundamental AI concepts and their practical applications. The challenge was to understand the complete workflow of an AI-based solution, beginning with data preparation and feature extraction and progressing toward model development and evaluation.

As a practical application of the concepts learned, the project component focused on developing a movie recommendation system capable of identifying movies similar to a selected movie based on its content and metadata.

4.4 Project Objectives

- To understand the fundamental concepts and applications of Artificial Intelligence.
- To develop basic programming and data-handling skills using Python.
- To learn techniques for data preprocessing, visualization, and preparation for AI applications.
- To understand the fundamentals of supervised learning and machine learning models.
- To learn the basics of model evaluation and performance measurement.
- To gain an introductory understanding of neural networks and deep learning.
- To understand the working and applications of Convolutional Neural Networks (CNNs) for image-related tasks.
- To explore Natural Language Processing and basic text-processing techniques.
- To understand the basic principles and applications of Reinforcement Learning and Q-learning.
- To apply the concepts learned during the internship through a practical AI project.

4.5 Scope of the Project

The scope of the internship covers the fundamental areas required to develop a basic understanding of Artificial Intelligence and its applications. It includes Python-based data handling, data preprocessing, supervised machine learning, model evaluation, neural networks, CNNs, NLP, and reinforcement learning.

The practical scope extends to implementing AI and machine learning concepts through basic examples and a project. The Content-Based Movie Recommendation

System demonstrates the application of data processing, textual feature representation, and similarity calculation to generate movie recommendations.

The internship provides a foundation for further exploration of advanced AI topics. Future work could include advanced deep learning architectures, larger datasets, more sophisticated NLP techniques, hybrid recommendation systems, and deployment of AI models as web or mobile applications.

4.6 Technologies and Tools Used

- **Python** – Used for programming and implementation of AI and machine learning concepts.
- **NumPy** – Used for numerical operations and array-based data processing.
- **Pandas** – Used for data manipulation, cleaning, and analysis.
- **Matplotlib** – Used for data visualization.
- **Scikit-learn** – Used for machine learning, feature extraction, similarity calculation, and evaluation.
- **NLTK and SpaCy** – Introduced for Natural Language Processing and text-processing tasks.
- **Jupyter Notebook / Google Colab** – Used for writing, executing, and documenting Python-based implementations.
- **Machine Learning techniques** – Used for understanding supervised learning and model evaluation.
- **Neural Networks and CNNs** – Studied as fundamental deep learning techniques.
- **Q-learning** – Studied as a basic Reinforcement Learning approach.

4.7 System Architecture

The overall internship consisted of multiple AI concepts and therefore did not follow a single system architecture. However, the practical project developed during the internship followed a data-processing and recommendation pipeline.

The architecture of the Content-Based Movie Recommendation System begins with a movie dataset containing information such as movie titles, genres, descriptions, keywords, and other metadata. The data is first loaded and preprocessed to handle missing or irrelevant information. Relevant attributes are then selected and transformed into a suitable feature representation.

The extracted features are converted into numerical vectors, which allow the system to mathematically compare movies. Similarity between movies is calculated using cosine similarity, after which the movies are ranked according to their similarity scores. Finally, the highest-ranked movies are presented as recommendations.

4.8 Methodology

The internship followed a progressive learning methodology, beginning with the fundamentals of Artificial Intelligence and Python programming and gradually moving toward more advanced AI concepts. The first stage focused on understanding AI concepts and using Python for data handling, followed by data preprocessing and visualization.

The second stage introduced machine learning concepts, particularly supervised learning and model evaluation. The internship then progressed to neural networks and deep learning, including the fundamentals of CNNs and their application to image classification. The final stage covered advanced applications such as NLP, text processing, sentiment analysis, and Reinforcement Learning using basic Q-learning.

Alongside the learning modules, practical implementation was used to reinforce the concepts. For the project component, a movie dataset was processed using Python and relevant movie features were extracted. The textual and categorical information was converted into a numerical representation, and cosine similarity was used to identify movies with similar content. The resulting similarity scores were used to generate a ranked list of recommendations.

4.9 Expected Outcomes

The expected outcome of the internship is a foundational understanding of Artificial Intelligence and its major areas of application. By completing the internship, the learner is expected to gain familiarity with Python-based data handling, preprocessing, machine learning, model evaluation, neural networks, CNNs, NLP, and reinforcement learning.

The internship also provides practical experience in applying these concepts to a real-world problem. Through the development of the Content-Based Movie Recommendation System, the learner gains experience in handling a dataset, processing textual information, extracting features, calculating similarity, and generating meaningful recommendations.

Overall, the internship is expected to provide a foundation for further learning and development in areas such as Machine Learning, Deep Learning, NLP, Computer Vision, and AI-based application development.

4.10 Certificates of Completion and Communication Proof

HR Offer-Letter of Internship

External

Inbox x



codectechnologies0@gmail.com

to me ▾

Sat, Aug 22, 8:14 PM



🏆 Welcome to Codec Technologies – Your Internship Journey Begins!

Dear Intern,

Congratulations and welcome to Codec Technologies! We're excited to have you on board as an intern and look forward to working with you. This is a meaningful opportunity, and we're eager to support your growth as you train with us.

Codec Technologies is dedicated to nurturing talent and fostering growth and innovation. During your internship, you'll gain hands-on experience, work on industry-level projects, and collaborate with our talented team. Completing all training lectures and projects is required to maximize your learning and progress.

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Assigned Guide Name -

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Designation - HR and Training Officer

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